

Superposition

Day 3

World building

Invisible Cities

Schedule

Day 3

On **Wednesday**, you will learn how to make your two systems interact and merge them into one unified, generative city.

9:15 - 9:45: Briefly present your installation sketch

9:45 - 10:45: Masterclass: Building worlds

10:45 - 11:00: Coffee break

11:00 - 11:45: Masterclass: Biomimicry

11:45 - 12:30: Building your node trees

12:30 - 14:00: Lunch

14:00 - 15:00: Materials, render modes & HDRI

15:00 - 17:00: Deep Work: Merging worlds

End-of-day: Print your screenshots and put them on the wall

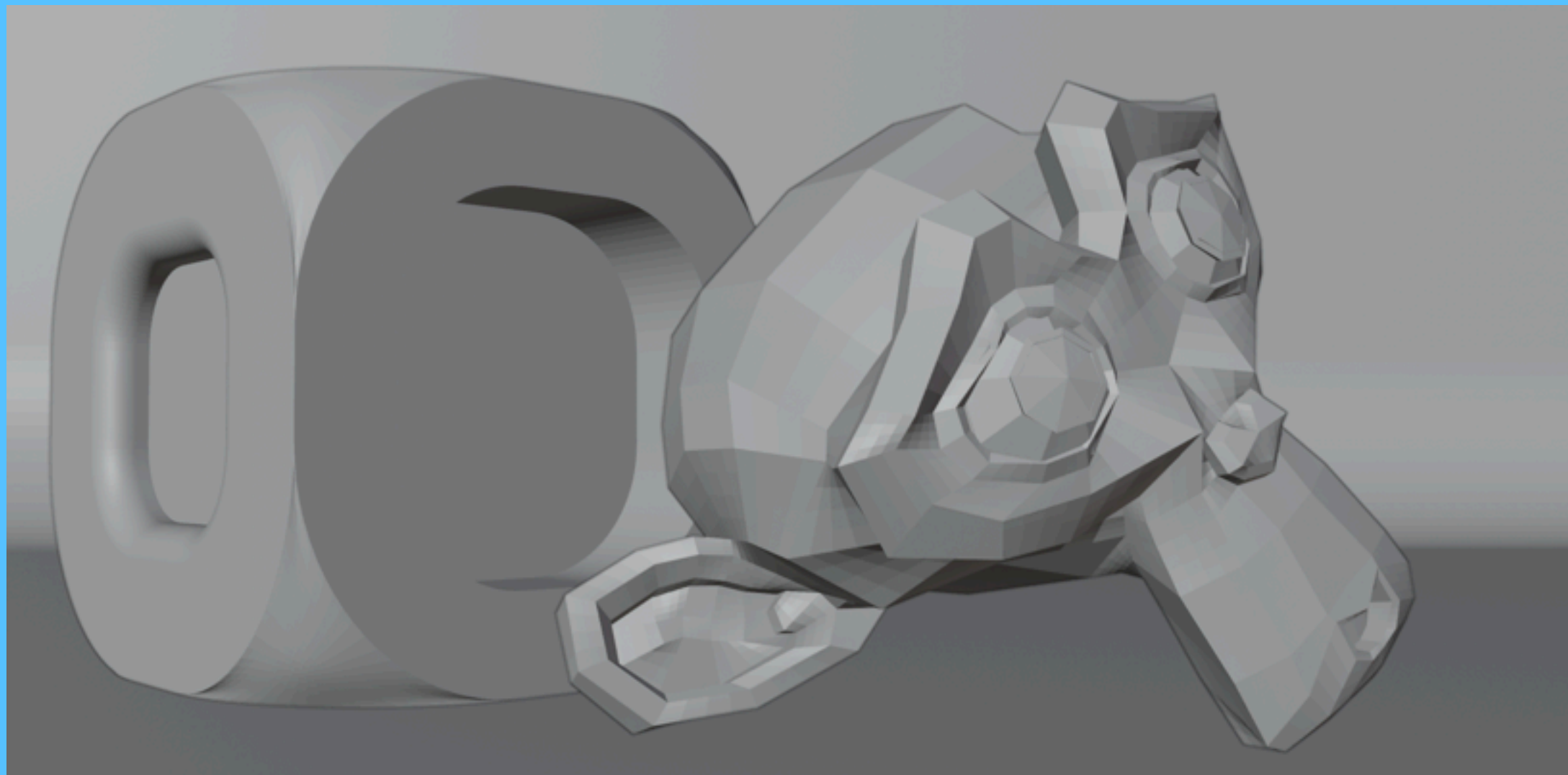
Glass in EEEVEE



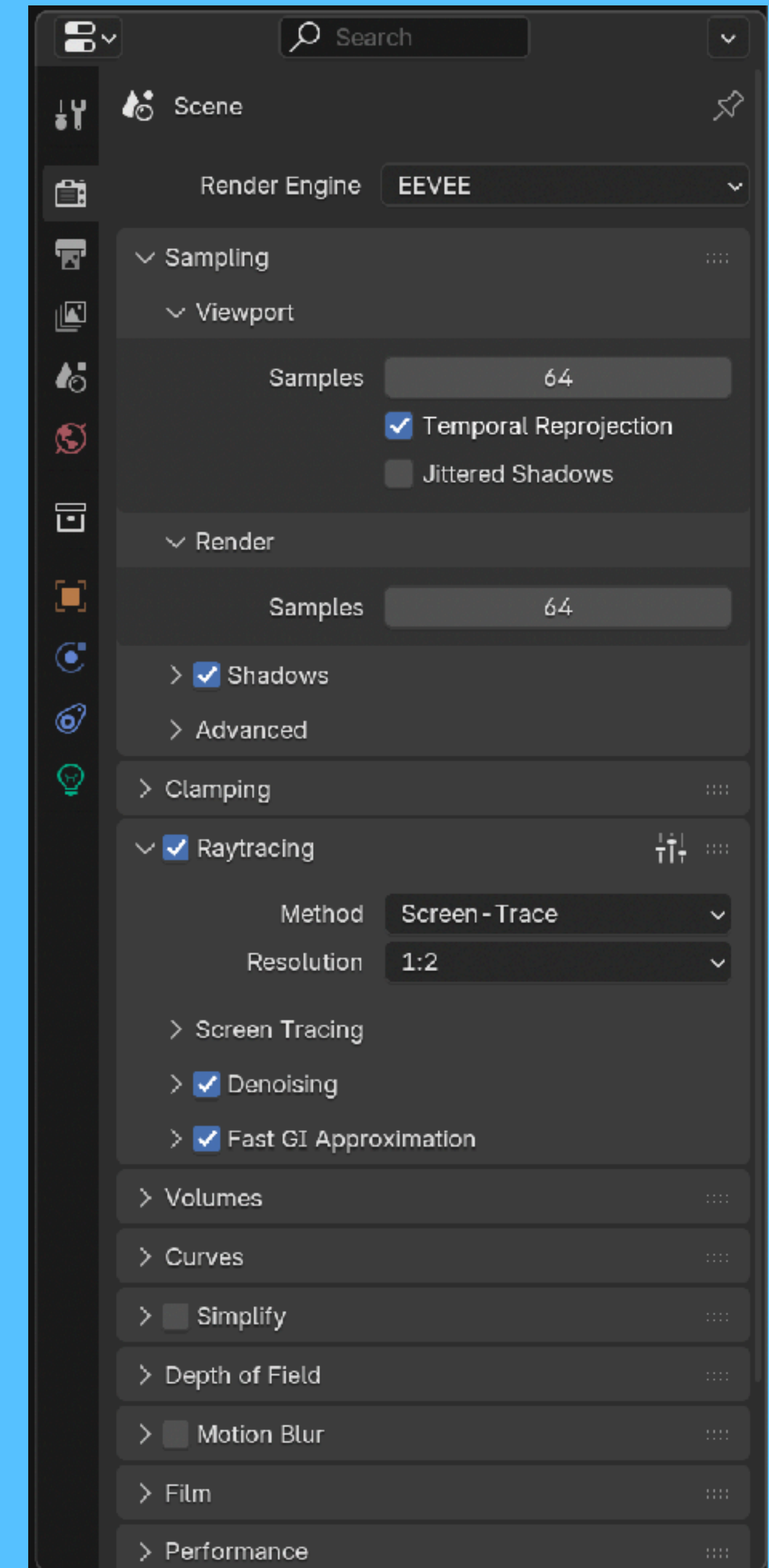
Invisible Cities

Setup your scene

1. Create a **new file**
2. **Open render properties**
3. **Increase the amount of samples**
4. **Turn on Raytracing**
5. **Add a backdrop and some meshes with a Thickness Modifier**



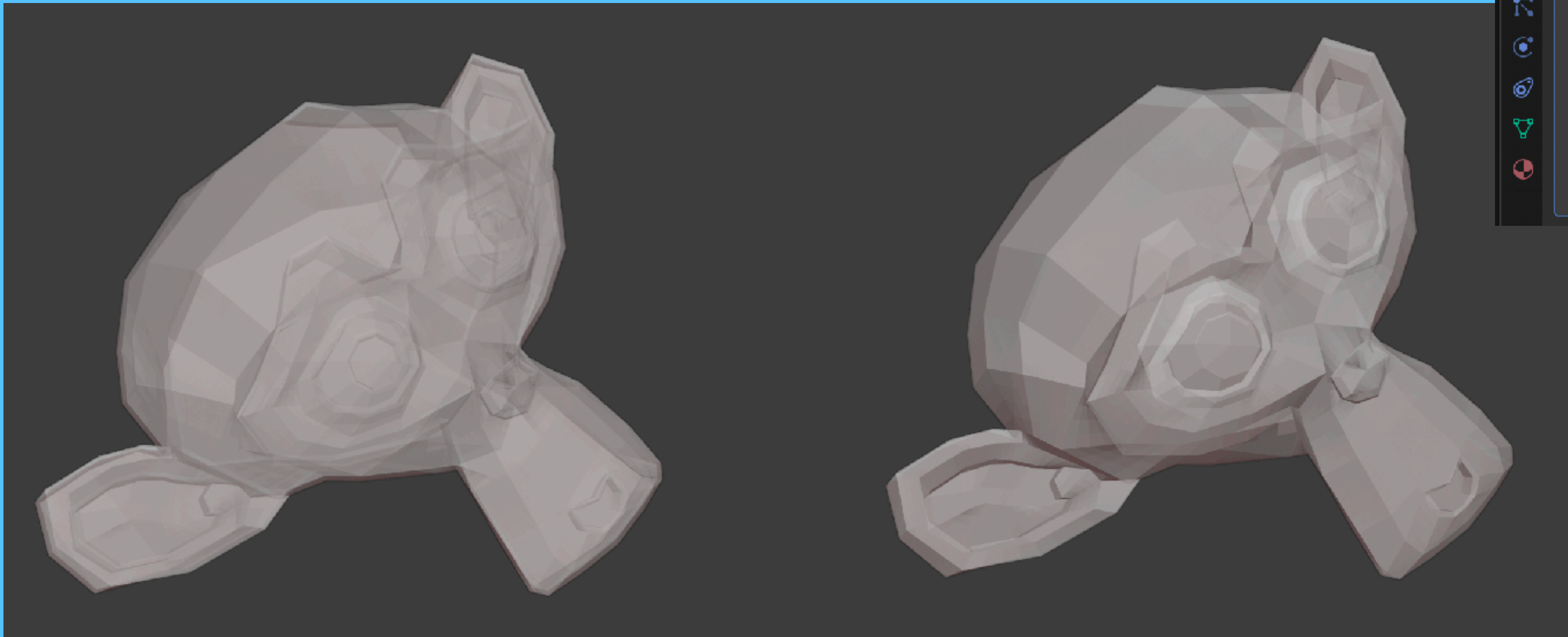
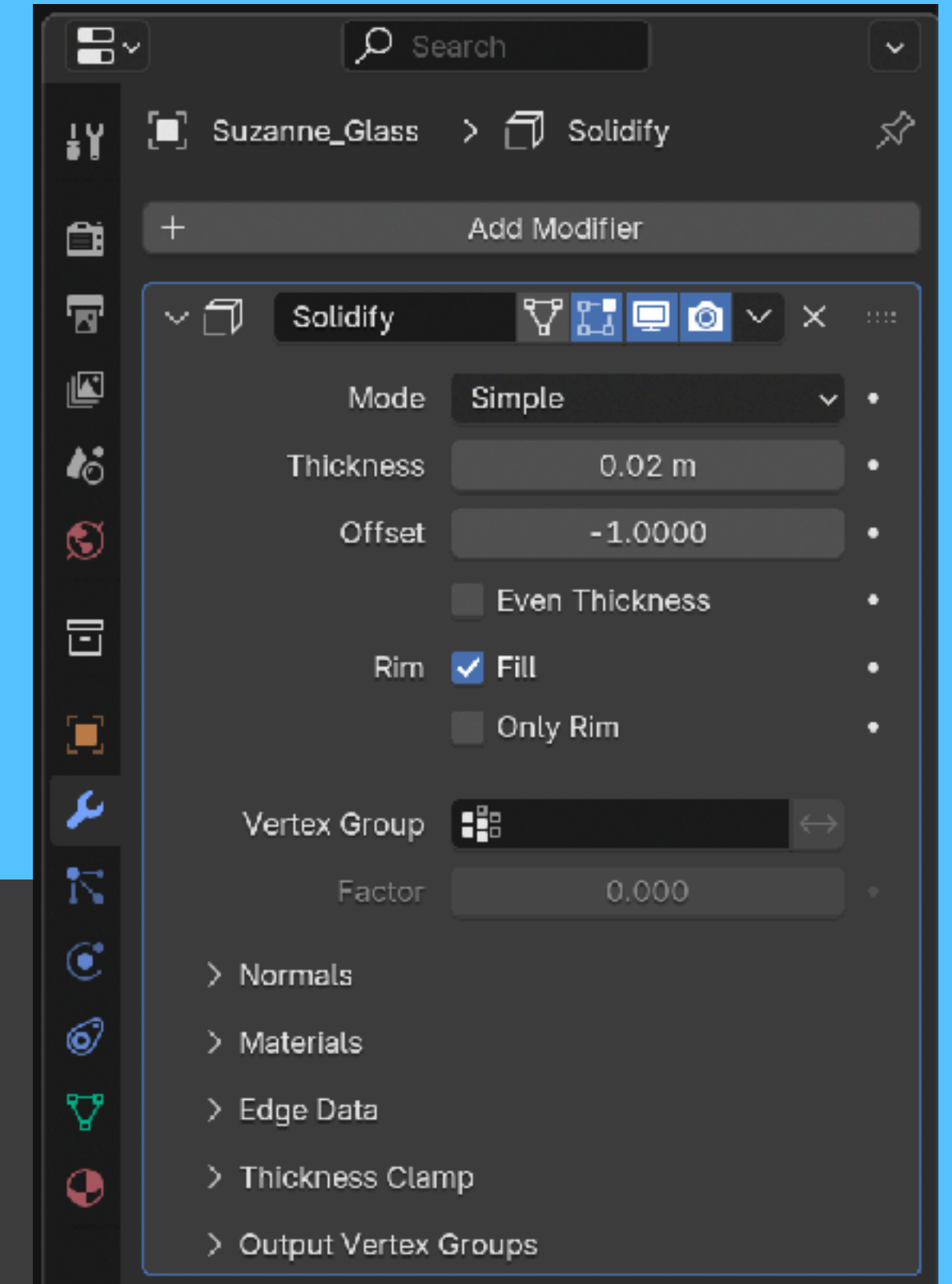
Glass



Setup your scene

6. Add a Solidify modifier in the Modifier Properties
7. Set the thickness to 0.02

The Solidify modifier takes the surface of any mesh and adds depth, thickness to it.



Setup your scene

8. Create a new material “Glass”
9. Set the Base Color to full white
10. Open Transmission and set Weight to 1.0
11. Almost there!

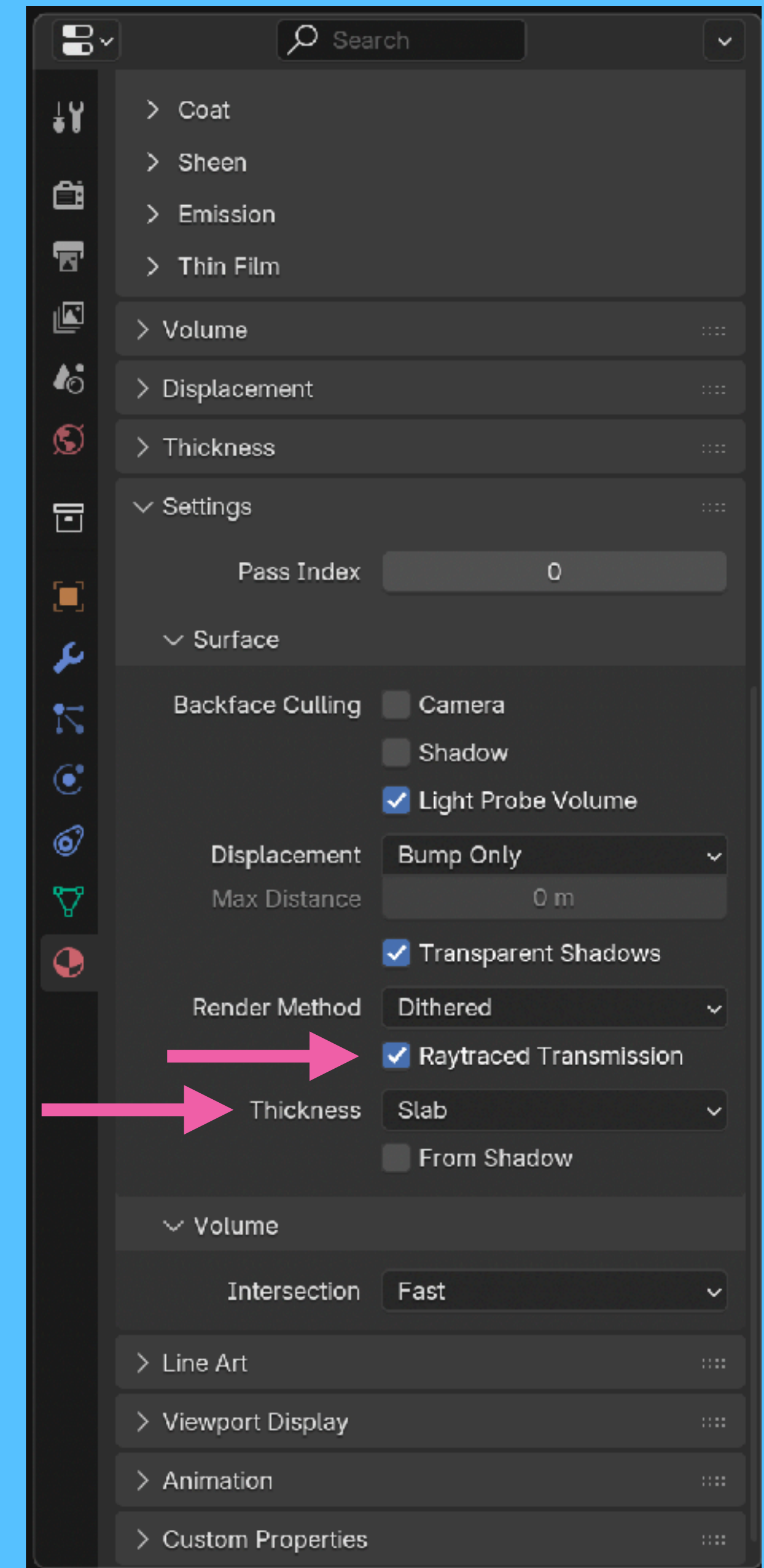


Glass



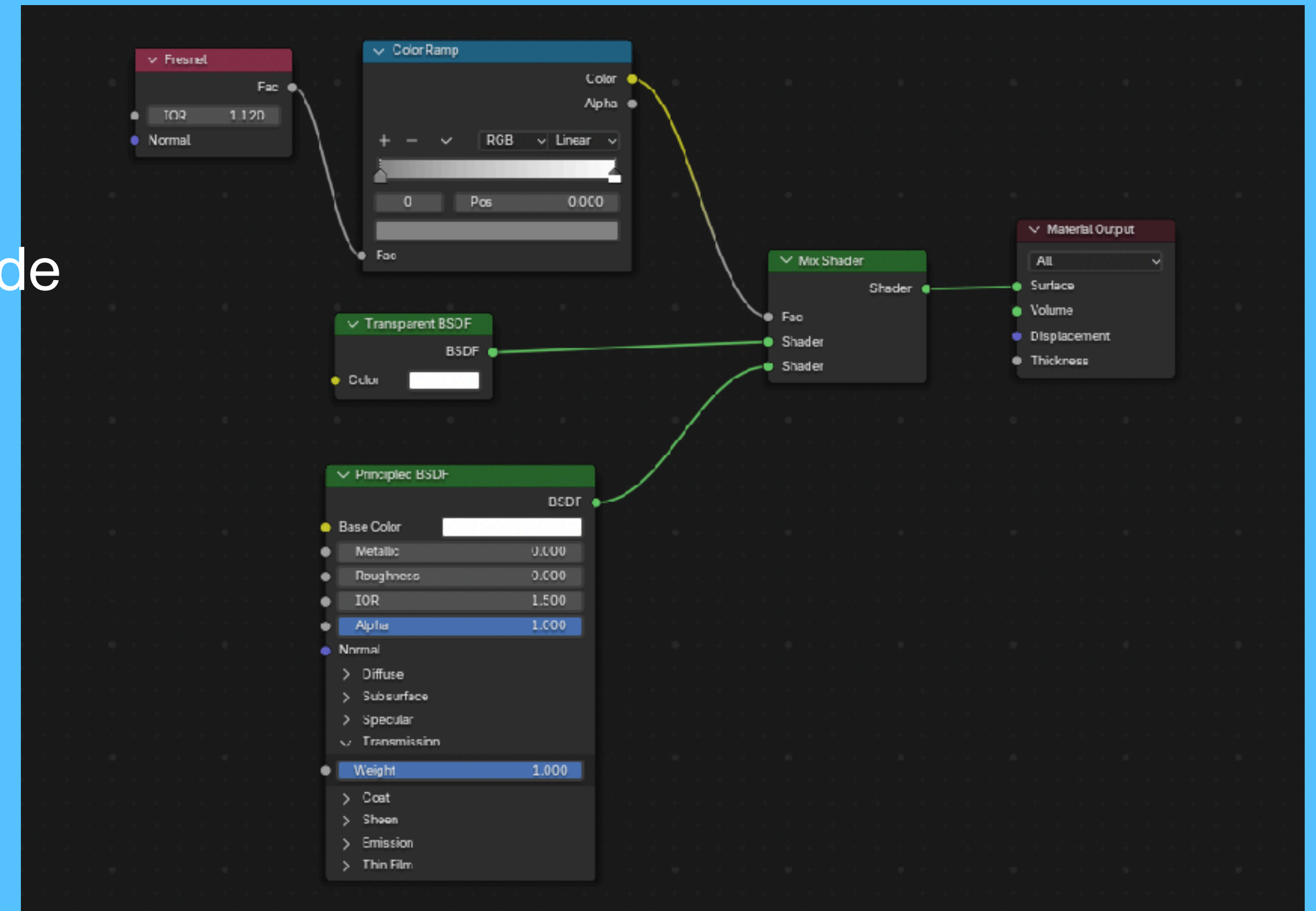
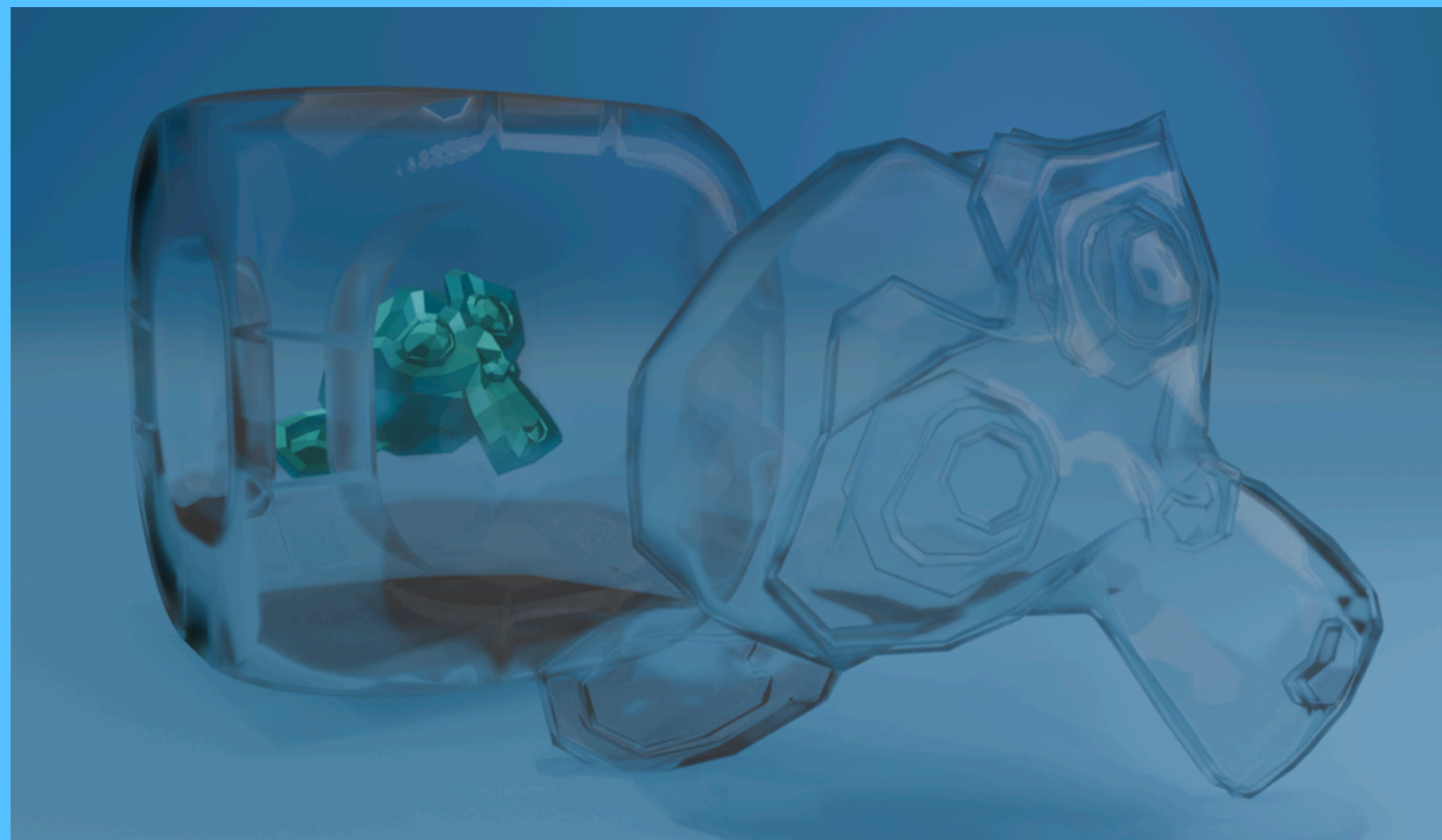
Setup your scene

12. With your glass object selected, open the Material Properties
13. **In Settings, change the following values:**
14. **Turn on Raytraced Transmission**
15. **Set Thickness to Slab**



Setup your scene

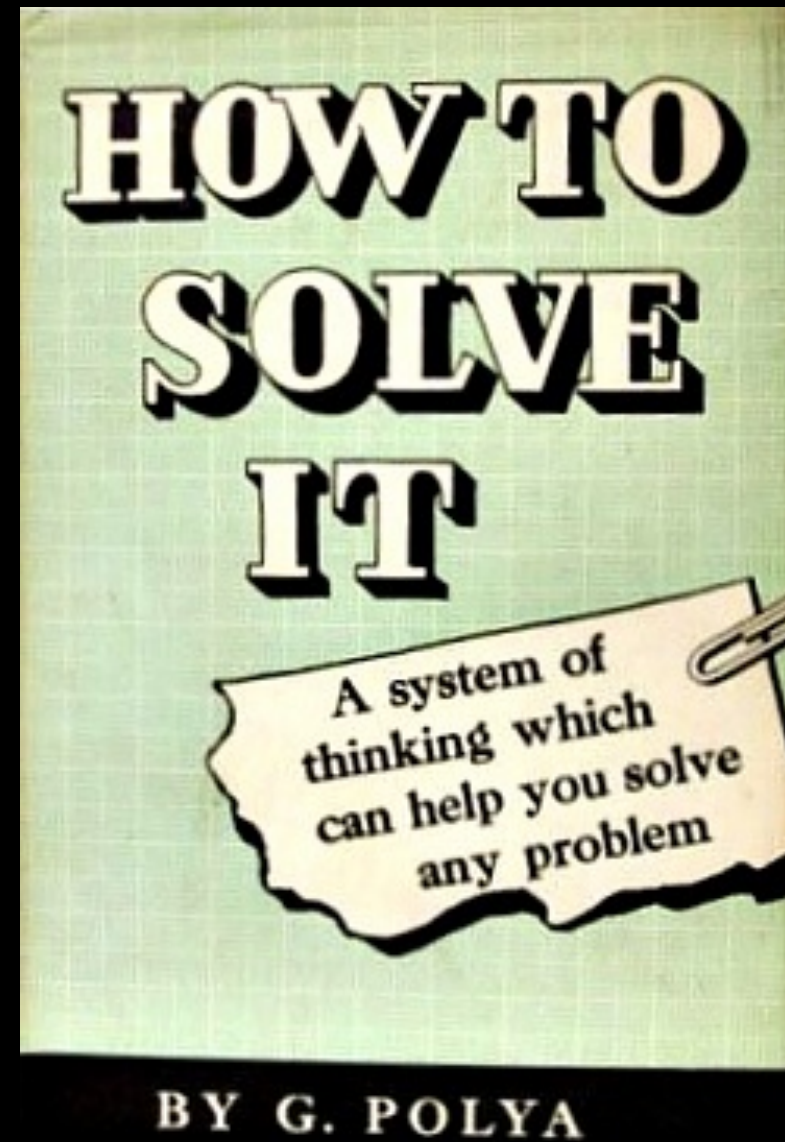
16. Open your glass material
17. Add a Fresnel node and set the IOR to 1.120
18. Add a Color Ramp and change black to grey
19. Add a Mix Shader node
20. Add a Transparent BSDF node
21. Mix the Principled and Transparent BSDF node



How to solve it?

Methods of problem solving
Reduce complexity to its simplest, small steps

How to solve it?



Four principles

1. First, you have to understand the problem
2. After understanding, make a plan
3. Carry out the plan
4. Look back on your work. How could it be better?

First principle

Understand the problem

1. What are you asked to find or show?
2. Can you restate the problem in your own words?
3. Can you think of a picture or a diagram that might help you understand the problem?
4. Is there enough information to enable you to find a solution?
5. Do you understand all the words used in stating the problem?
6. Do you need to ask a question to get the answer?

Second principle

Devise a plan

There are many reasonable ways to solve problems, e.g.:

1. Guess and check
 2. Eliminate possibilities
 3. Make an orderly list
 4. Look for a pattern
 5. Draw a picture
 6. Solve a simpler problem
 7. Use a model
 8. Work backward
 9. Use a formula
 10. Be creative
- [...]

Third principle

Carry out the plan

This step is usually easier than devising the plan. In general, all you need is care and patience, given that you have the necessary skills. Persist with the plan that you have chosen. If it continues not to work, discard it and choose another. Don't be misled; this is how mathematics is done, even by professionals.

Fourth principle

Review / extend

Much can be gained by taking the time to reflect and look back at what you have done, what worked and what did not, and with thinking about other problems where this could be useful. Doing this will enable you to predict what strategy to use to solve future problems, if these relate to the original problem.